Parallel Computation

1 Complexity

1.1 Time Complexity of a Decision Problem $O_T[n]$

Define Time Complexity $O_T[n]$ of solution s^+ to Decision Problem D as the total number of logical operations

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_i] := P :$$

$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \;\; \cap \;\; (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$

$$s^+ = \{s_1, s_2, ..., s_N | b_1, b_2, ..., b_M, y_o\} = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\}$$
$$= \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

1.2 Space Complexity $O_S[n]$

Define Space Complexity $O_S[n]$ of solution s^+ to Decision Problem D as the total number of memory elements

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_i] := P :$$

$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \;\; \cap \;\; (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$

$$s^+ = \{s_1, s_2, ..., s_N | b_1, b_2, ..., b_M, y_o\} = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\}$$
$$= \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_S[n] := |\mathcal{M}| + |y_o|^* = M + 1$$

*It is convention to reserve one memory element for output y_o.
Void programs do not require the y_o memory element for output

2 Definition of Complexity

Define Complexity $O[n]$ as a vector of dimension V

$$\mathbf{O}[n] := \ < O_T[n], O_S[n], O_3[n], O_4[n]..., O_V[n] >$$

3 Simple Computational Complexity

The remainder of this document assumes simple computational complexity of dimension 2

3.1 Definition

Define simple computational complexity of dimension 2

$$\mathbf{O}[n] := \ < O_T[n], O_S[n] >$$

3.2 Total Complexity

Define Total Complexity of solution s^+

$$O[n] := |s^+[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o| = N + M + 1$$

3.3 Time Complexity

Restate definition of Time Complexity $O_T[n]$ of solution s^+

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

3.4 Space Complexity

Restate definition of Time Complexity $O_S[n]$ of solution s^+

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_S[n] := |\mathcal{M}| + |y_o| = M + 1$$

3.5 Total Complexity as a Function of Time and Space Complexity

$$O[n] := |s^+[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o|$$

$$= O_T[n] + O_S[n]$$

3.6 $O_S[n] > 0^*$

Assuming Program is not void

3.6.1 Proof

Assume $O_S[n] = 0$

$$O_S[n] := |\mathcal{M}| + |y_o|$$
$$O_S[n] = 0 \Rightarrow \mathcal{M} = y_o = \varnothing$$
$$y_o = \varnothing; \;\; y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_S[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_S[n] \geqslant 0 \text{ by definition of magnitude}$$
$$\therefore O_S[n] > 0$$

3.7 $O_T[n] > 0^*$

Assuming Program is not void

3.7.1 Proof

Assume $O_T[n] = 0$

$$O_T[n] := |\mathcal{L}|$$
$$O_T[n] = 0 \Rightarrow y_o \notin \{\mathbb{T}, \mathbb{F}\}$$
$$y_o \notin \{\mathbb{T}, \mathbb{F}\}; \;\; y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_T[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_T[n] \geqslant 0 \text{ by definition of magnitude}$$
$$\therefore O_T[n] > 0$$

3.8 $O[n] > 0^*$

Assuming Program is not void

3.8.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0; \;\; O_S[n] > 0$$

$$\therefore O[n] > 0$$

3.9 $O[n] > O_T[n]_*$

Assuming Program is not void

3.9.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_S[n] > 0$$

$$\therefore O[n] > O_T[n]$$

3.10 $O[n] > O_S[n]^*$

Assuming Program is not void

3.10.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0$$

$$\therefore O[n] > O_S[n]$$

3.11 $O[n+1] \geqslant O[n]$

3.11.1 Proof

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$O[n] = \mid s^+[X_i] \mid$$

$$O[n+1] = \hat{O}[n] = \mid s^+[\hat{X}_i] \mid$$

For general solutions s^+

$$s^+[\hat{X}_i] \supseteq s^+[X_i]$$
$$\Rightarrow |s^+[\hat{X}_i]| \geqslant |s^+[X_i]|$$
$$\therefore \hat{O}[n] = O[n+1] \geqslant O[n]$$

4 Complexity of Canonical Instructions

$$c := a \leftarrow l[X_n]$$

5 Complexity of Computational Operations

5.1 $+$

Express the bounds of complexity for Computational Operation $+$

6 Divergent Complexity

6.1 Definition of Converges to

$$f[n] \textit{ converges to } C = \textit{convergent}[f[n], C] = a_o;\ a_o \in \{\mathbb{T}, \mathbb{F}\} =$$

$$|C - f[n+1]| < |C - f[n]| \ \ \forall n$$
$$\wedge$$
$$\nexists K : |C - f[\hat{n}]| > K \quad \forall n; K > 0$$

6.1.1 Notation

C is commonly denoted by a limit

$$C = lim_{n \to \infty} f[n]$$

6.2 Definition of General Convergence

$$f[n] \text{ is } \textit{convergent} = \textit{convergent}[f[n]] = a_o;\ a_o \in \{\mathbb{T}, \mathbb{F}\} =$$

$$\exists C :$$
$$\textit{convergent}[f[n], C] == \mathbb{T}$$

Alternatively

$$f[n] \text{ is } convergent = convergent[f[n]] = a_o \in \{\mathbb{T}, \mathbb{F}\} =$$

$$\exists C :$$

$$f[n] \ converges \ to \ C$$

6.3 Definition of Divergence

$$diverges[f[n]] = \neg converges[f[n]] = d_o; \ d_o \in \{\mathbb{T}, \mathbb{F}\}$$

$$:= \nexists C : convergent[f[n], C] == \mathbb{T}$$

6.4 Alternate Definition of Divergence

$$diverges[f[n]] = \neg converges[f[n]] = d_o; \ d_o \in \{\mathbb{T}, \mathbb{F}\}$$

$$= convergent[f[n], C] == \mathbb{F} \quad \forall C$$

6.4.1 Proof of Equivalence; Alternate Definition of Divergence

6.5 ?Necessary or Sufficient? Criteria 1 For Divergence

? The derivative as a function of K ? Function f[n] diverges if the Kth derivative of f[n] is strictly increasing

$$diverges[f[n]] := \nexists C : convergent[f[n], C] == \mathbb{T}$$

$$\Longleftrightarrow$$

$$\Delta_n^{K+1} f[n] > \Delta_n^K f[n] \quad \forall K$$

Alternatively

$$\Delta_n^{K+1} f[n] - \Delta_n^K f[n] > 0 \quad \forall K$$

$$\Delta_n^{K+2} > 0 \ \ \forall K$$

6.5.1 Criteria 1; Proof of Necessity and Sufficiency

$$diverges[f[n]] = d_o; \ d_o \in \{\mathbb{T}, \mathbb{F}\}$$

$$= \nexists C : convergent[f[n], C] == \mathbb{T}$$

Let

$$f[n]:$$
$$\Delta_n^{K+2} > 0 \quad \forall K$$

6.6 ?Necessary or Sufficient? Criteria 2 For Divergence

Function f[n] diverges if the Derivative as a function of K does not Converge

$$f[n] \; is \; Divergent = Divergent[f[n]] = a_o \in \{\mathbb{T}, \mathbb{F}\} :=$$

$$\nexists c : lim_{n\to\infty} \Delta_n^K f[n] = c$$

6.6.1 Criteria 2; Proof of Necessity and Sufficiency

$$diverges[f[n]] = d_o; \; d_o \in \{\mathbb{T}, \mathbb{F}\}$$

$$= \nexists C : convergent[f[n], C] == \mathbb{T}$$

6.7 Verbal Expressions

$$f[n] \; diverges = f[n] \; is \; divergent =$$
$$f[n] \; is \; not \; convergent = f[n] \; does \; not \; converge$$

6.8 Definition of Divergent Function

6.8.1 Definition 1

Define Divergent Function f[n] having strictly increasing K^{th} derivative

$$f[n] \; is \; Divergent = Divergent[f[n]] = a_o \in \{\mathbb{T}, \mathbb{F}\} :=$$

$$\Delta_n^{K+1} f[n] > \Delta_n^K f[n] \quad \forall K$$

Alternatively

$$\Delta_n^{K+1} f[n] - \Delta_n^K f[n] > 0 \quad \forall K$$

$$\Delta_n^{K+2} > 0 \;\; \forall K$$

6.8.2 Definition 2

$$f[n] \; is \; Divergent = Divergent[f[n]] = a_o \in \{\mathbb{T}, \mathbb{F}\} :=$$

$$\nexists c : lim_{n\to\infty} \Delta_n^K f[n] = c$$

6.8.3 Proof of Equivalence Definition 1 $\Leftrightarrow$ Definition 2

6.8.4 Sufficient Proof

$$f[n] \; is \; Divergent \iff$$

$$\nexists c : lim_{n\to\infty} \Delta_n^K f[n] = c$$

6.8.5 Necessary Proof

$$f[n] \; is \; Divergent \iff$$

$$\nexists c : lim_{n\to\infty} \Delta_n^K f[n] = c$$

Proof by contradiction of definition of limit Using the definition of increasing convergence for a discrete function*

6.9 Defintion

Decision problem $\hat{D}$ with solution s^+ has divergent total complexity $O[n]$ if

$$lim_{n\to\infty} \frac{O[n+1]}{O[n]} \;\; diverges$$

6.10 Divergent Problems

$$\hat{\mathcal{D}} := \{\hat{D}_1, \hat{D}_2, ...\} :$$
$$lim_{n\to\infty} \frac{O[n+1]}{O[n]} \;\; diverges \quad \forall s^+ \in S_i^+, \; \hat{D}_i \in \hat{\mathcal{D}}$$

6.11 The Set of Polynomial Solutions and the Set of Divergent Solutions are disjoint

$$\mathbb{P} \cap \hat{D} = \varnothing$$

6.12 Proof

Let D $\in \hat{\mathcal{D}}$

$$lim_{n\to\infty} \frac{O[n+1]}{O[n]} \;\; diverges \text{ by definition}$$

Assume D $\in \mathbb{P}$

$$lim_{n\to\infty}\frac{O[n+1]}{O[n]} = 1$$

$$lim_{n\to\infty}\frac{O[n+1]}{O[n]} = 1 \text{ contradicts the definition of Divergent Problems}$$

$$\therefore D \in \hat{\mathcal{D}} \Rightarrow D \notin \mathbb{P}$$

Let D $\in \mathbb{P}$

$$lim_{n\to\infty}\frac{O[n+1]}{O[n]} = 1 \text{ by property of Polynomial complexity}$$

Assume D $\in \hat{D}$

$$lim_{n\to\infty}\frac{O[n+1]}{O[n]} \text{ diverges}$$

$$lim_{n\to\infty}\frac{O[n+1]}{O[n]} \text{ diverges contradicts a property of Polynomial complexity}$$

$$\therefore D \in \mathbb{P} \Rightarrow D \notin \hat{\mathcal{D}}$$

$$\therefore \mathbb{P} \cap \hat{\mathcal{D}} = \varnothing$$

7 Workers

8 Definition of Parallel Programs

9 Can Parallel Workers transform a Divergent task into a Polynomial Task

10 Conservation of Computation

11 Sum to N Problem with 2 integers

11.1 State formal definition of Sum to N : $x_i + x_j == N$

$$X_n = \{x_1, ..., x_n\}$$

$$D := f[X_i, N] = a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_n] = P[X_n] :$$
$$(P[X_i] = y_o == a_o \quad \forall X_i) \quad \cap \quad (P[X_{n+1}] \supseteq P[X_n] \quad \forall X_{n+1})$$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$D = f[X_i] = \quad \exists x_j, x_k \in X_n \quad j \neq k :$$
$$x_j + x_k == N$$

11.2 Express a formal solution : $O_S[n] \sim n^0$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i < n \ , \ n \geqslant j > i$$
$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$
$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$

11.3 Prove s^+ satisfies the subfunction condition of solutions: $P[X_{n+1}] \supseteq P[X_n] \quad \forall X_{n+1}$

$$X_n = \{x_1, x_2, ..., x_n\}; \quad X_{n+1} = \{x_1, x_2, ..., x_n, x_{n+1}\}$$
$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s^+_{n+1} = s^+ \cup \hat{s}^+$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i < n \ , \ n \geqslant j > i$$

$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$

$$\forall k < n+1$$
$$s... = b_1 \leftarrow x_k + x_{n+1}$$
$$s... = b_1 \leftarrow b_1 == N$$
$$s... = y_o \leftarrow y_o \vee b_1$$

$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$
$$\hat{s}^+ = \{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$
$$s^+_{n+1} = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\} \ \cup$$
$$\{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$

$$s^+_{n+1} = s^+ \cup \hat{s}^+ = P[X_{n+1}] \supseteq P[X_n] = s^+$$

11.4 Determine $O[n], O_S[n], O_T[n], f_{n+1}[n], f^T_{n+1}[n], f^S_{n+1}[n]$ for the above solution

$$O_S[n] = |y_o| + |b_1| = 2$$
$$O_T[n] = 3n(n-1) + 1 = 3n(n-1) - 1 + O_S[n]$$
$$O[n] = 3n(n-1) + 3 = 3n^2 - 3n + 3$$

$$f^S_{n+1}[n] = 0$$
$$f^T_{n+1}[n] = 6n$$
$$f_{n+1}[n] = f^S_{n+1}[n] + f^T_{n+1}[n]$$

11.5 Verify $O[n+1] = O[n] + f_{n+1}[n]$

$$O[n+1] = O[n] + \hat{O}[n]$$
$$3(n+1)^2 - 3(n+1) + 3 = 3n^2 - 3n + 3 + 6n$$
$$3n^2 + 6n + 3 - 3n - 3 + 3 = 3n^2 + 3n + 3$$

$$3n^2 + 3n + 3 = 3n^2 + 3n + 3$$

11.6 Show s^+ has Polynomial Complexity by the definition of Total Polynomial Complexity

$$O[n] = 3n^2 - 3n + 3$$

11.7 Show the limit $_{n\to\infty}\frac{O[n+1]}{O[n]}$ does not Diverge

$$limit_{n\to\infty}\frac{O[n+1]}{O[n]} =$$

$$limit_{n\to\infty}\frac{3n^2+3n+3}{3n^2-3n+3} =$$

$$limit_{n\to\infty}(\frac{3n^2-3n+3}{3n^2-3n+3} + \frac{6n}{3n^2-3n+3}) =$$

$$limit_{n\to\infty}(1 + \frac{6n}{3n^2-3n+3}) = 1$$

12 The Knapsack Problem

12.1 The Knapsack Problem

The Knapsack Problem is a famous problem in computer science which asks if objects can be stored in a knapsack. Typically the problem is designed with two constraints, weight and value. Given objects x_i, each with a respective weight w_i and value v_i, does there exist a combination of objects lighter than input weight W and more valuable than input value V?

12.2 Formal Definition

$$X_n = \{x_1, x_2, ..., x_n\} = \{\{w_1, v_1\}, \{w_2, v_2\}, ..., \{w_n, v_n\}\}$$

$$I = \{i_1, i_2, ..., i_n\} : i_l \in \{0, 1\} \ \ \forall i_l \in I$$

$$D := f[X_n, W, V] = a_o \in \{\mathbb{T}, \mathbb{F}\} = \exists I :$$
$$(\textstyle\sum_{j=1}^{n} i_j w_j < W) \wedge (\sum_{j=1}^{n} i_j v_j \geqslant V)$$

12.3 Solve for C[n]

12.3.1 Expressing I as a binary number

$$I = \{i_1, i_2, ..., i_n\} : i_l \in \{0, 1\} \ \ \forall i_l \in I$$

Valid combinations of I

$$I_{valid} =$$
$$\{\{0, 0, 0, ..., 0, 0, 1\}, \{0, 0, 0, ..., 0, 1, 0\}, \{0, 0, 0, ..., 0, 1, 1\}, ..., \{1, 1, 1, ..., 1, 1, 1\}\}$$

$$C[n] = |I_{valid}[n]| = 2^n - 1$$

12.3.2 Using a sum of combinations of inputs x_i

$$X_n = \{x_1, x_2, ..., x_n\} = \{\{w_1, v_1\}, \{w_2, v_2\}, ..., \{w_n, v_n\}\}$$

Valid combinations of x_i

$$X_{valid}[n] =$$

$$\{x_1\}\cup\{x_2\}\cup...\cup\{x_n\}\cup\{x_1,x_2\}\cup\{x_1,x_3\}\cup...\cup\{x_{n-1},x_n\}\cup...\cup\{x_1,x_2,...,x_n\}$$

$$= \ {}_{X_n}C_1 \cup \ {}_{X_n}C_2 \cup ... \cup \ {}_{X_n}C_n$$

$$C[n] = |X_{valid}[n]| = \sum_{j=1}^{n} \ {}_nC_j$$

12.3.3 Verify consistency

$$C[n] = |X_{valid}[n]| = |I_{valid}[n]|$$

$$= 2^n - 1 = \sum_{j=1}^{n} \ {}_nC_j = {}_nC_1 + {}_nC_2 + ... + {}_nC_n$$

$$= 2^n - 1 = 2^n - 1$$

12.4 Express a solution s^+ of the Knapsack Problem

12.5 Prove s^+ satisfies the subfunction condition of solutions

12.6 Determine $O[n], O_T[n], O_S[n], f_{n+1}[n]$

12.7 Show $s^+ \notin \mathbb{P}$

12.8 Express the Solution Space $\mathbb{S}$ for The Knapsack Problem

12.9 Prove a lower bound for all solutions $s^+ \in S^+ := O_{lower}[n]$

12.10 Prove D has Divergent Complexity

Appendix

13 Criticism of Overloaded Equivalence

In computer science, it is convention to overload equivalence =

$$a_i = a_i \quad \forall a_i \in \Omega$$

Consider standard C++ syntax
int x = 3;
int y = 4;
int z = x + y;

Int x is not inherently equal to 3. Rather, we are creating an open space "x" for a value and setting the value to 3. Similarly, z is not inherently equal to the value of x + y. Rather, we are creating an open space "z" for a value and setting the value to the sum of x and y which have already been set.

$$x \leftarrow 3$$

$$y \leftarrow 4$$

$$z \leftarrow x + y$$

14 Existence of $O_{max}[n]$

14.1 Proof

14.1.1 Alternate Definition; Left Hand Derivative

Some sources define

$$\Delta_n^1 f[n] = f[n] - f[n-1]$$

Citations

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